

James Gardner

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Education

Lancaster University

Oct 2021 – July 2025

Master of Science (MSci) in Computer Science – First Class Honours Degree

Skills

Languages: Java, Python, C#, JavaScript, SQL, HTML, CSS

Technologies: Git, CI/CD (GitHub Actions), Linux, Docker, Kubernetes, Azure

Experience

Software Engineer Intern, Lancashire Teaching Hospitals NHS Trust

Jan 2025 – June 2025

- Contributed to the continuous development of a secure NHS research platform using Python, Docker, Kubernetes, Azure and GitOps workflows.
- Created a containerised research environment enabling researchers to run and fine-tune pre-trained Large Language Models (LLMs) incorporating feedback from end users.
- Developed a FastAPI-based server compliant with the Model Context Protocol to expose structured tool functionality to local LLM agents within the research environment.

Projects

Distributed Task Submission System

jgardner01.github.io/Chord-DHT

- Designed and implemented a fault-tolerant distributed task submission platform using Java RMI and the Chord DHT protocol supporting passive replication.
- Built a custom REST server enabling users to submit tasks through a browser interface.
- Tasks are distributed and processed asynchronously by the appropriate Chord node with results retrievable via dynamically generated XML endpoints.

WebRTC Peer-to-Peer Communication Platform

jgardner01.github.io/WebRTC-Comm

- Built a browser-based real-time communication system using WebRTC, Node.js and Express for encrypted peer-to-peer voice, video, text communication and file sharing.
- Developed a socket-based signalling server to establish WebRTC connections and manage devices within a user-created virtual network.

URL Shortener

jgardner01.github.io/URL-Shortener

- Developed a full-stack URL shortening platform using Python, Flask, MongoDB and Google Safe Browsing API, featuring customisable short link generation, password protection, QR code generation, expiration rules and browser session/account-based tracking.
- Built a responsive dashboard with Bootstrap for link management and monitoring across sessions and devices.

Additional Work Experience

Store Assistant, Tesco Northampton South Extra

Nov 2020 – Present

- Maintained stock levels, visual standards and product mapping primarily in the beer, wine and spirits department with additional responsibilities in grocery and warehouse operations.
- Trained and supported new colleagues by explaining procedures and helping them adjust during their first weeks.
- Interacted with customers to resolve queries, provide product guidance and ensure a positive shopping experience.

Interests

Strength Training, Lancaster University Boxing Society, Travelling